

Subzero

Usability Test Report



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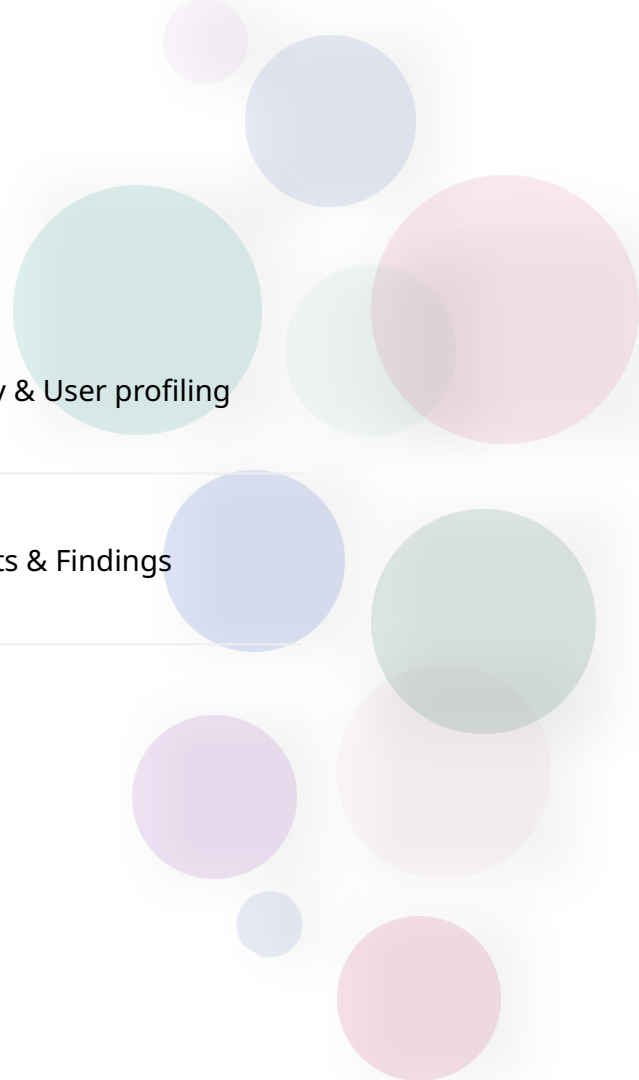
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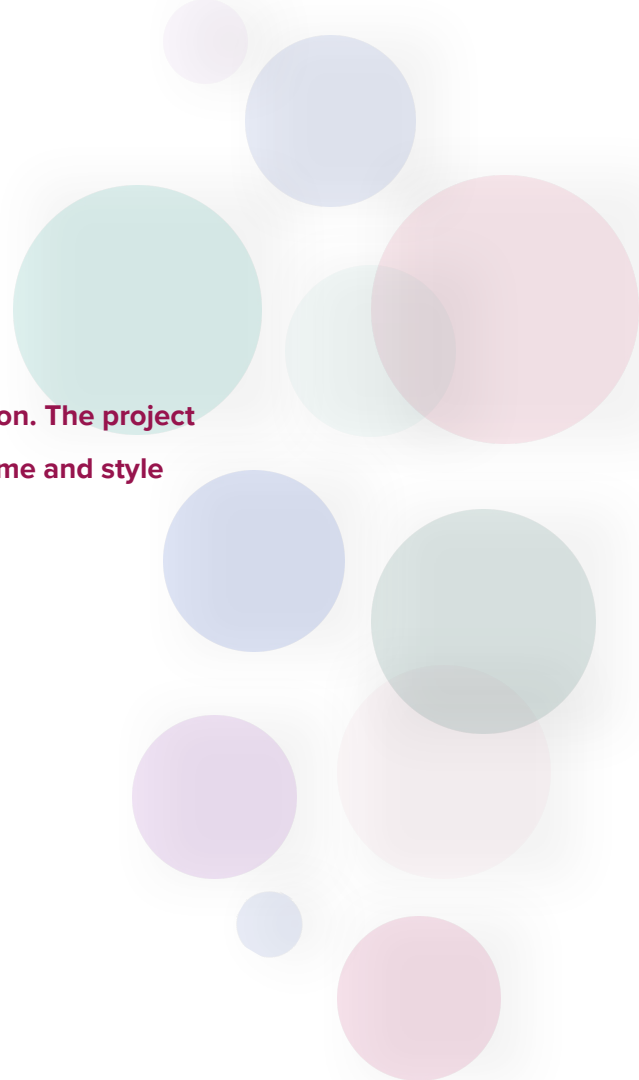
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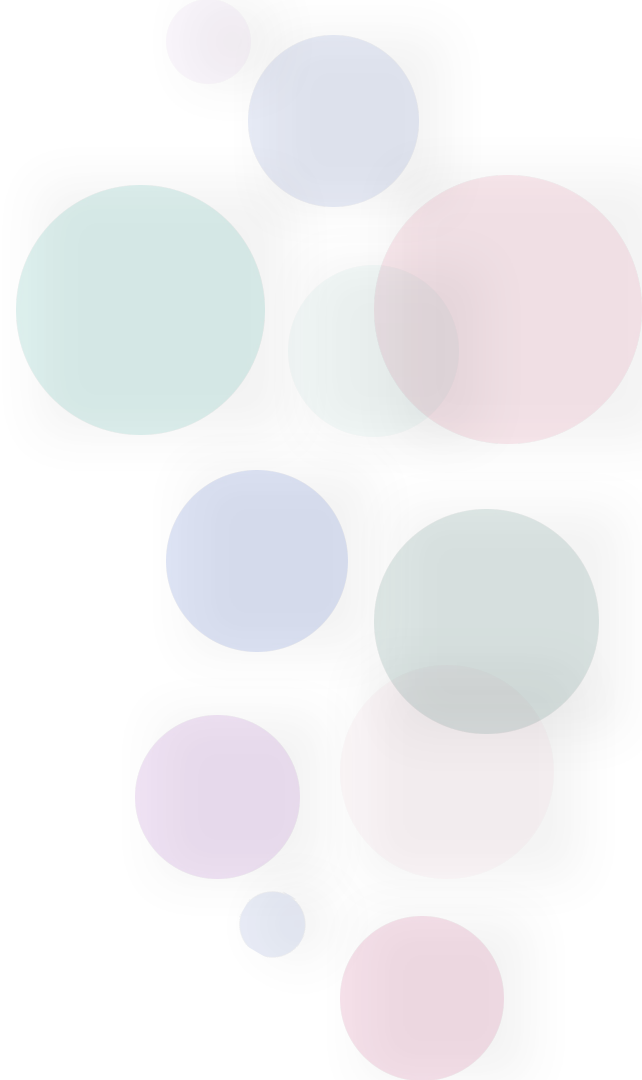
Overview

The Subzero design team has created a new color palette of 7 colors for data visualization. The project aims to explore and test the inclusion of data visualization colors that align with our theme and style while considering color blindness concerns.



Objective

- To ensure that all users with varying degrees of **color blindness may readily distinguish and comprehend the colors**
- To check the **chosen color for the UI elements will improve the user experience**
- To understand the **pinpoints or gray areas that the users with colorblindness face.**



Methodology and User profiling

In-depth Interview: Focused 45 min- 1 hour interviews with users to test the prototype screens

Cohorts-

Users who have color blindness

- a. Protanopia
- b. Treutenopia
- c. Deutanopia

Sample size: 12-16 (6-8 Axis employee users and 6-8 external users)

Age profile- 25 - 45

No	Name	User	color blind
1	User 1	DBAT	Deutanopia
2	User 2	External	Protanopia
3	User 3	DBAT	Protanopia
4	User 4	External	Deutanopia
5	User 5	External	Protanopia
6	User 6	External	Deutanopia
7	User 7	External	Tritanopia
8	User 8	External	Deutanopia
9	User 9	External	Deutanopia

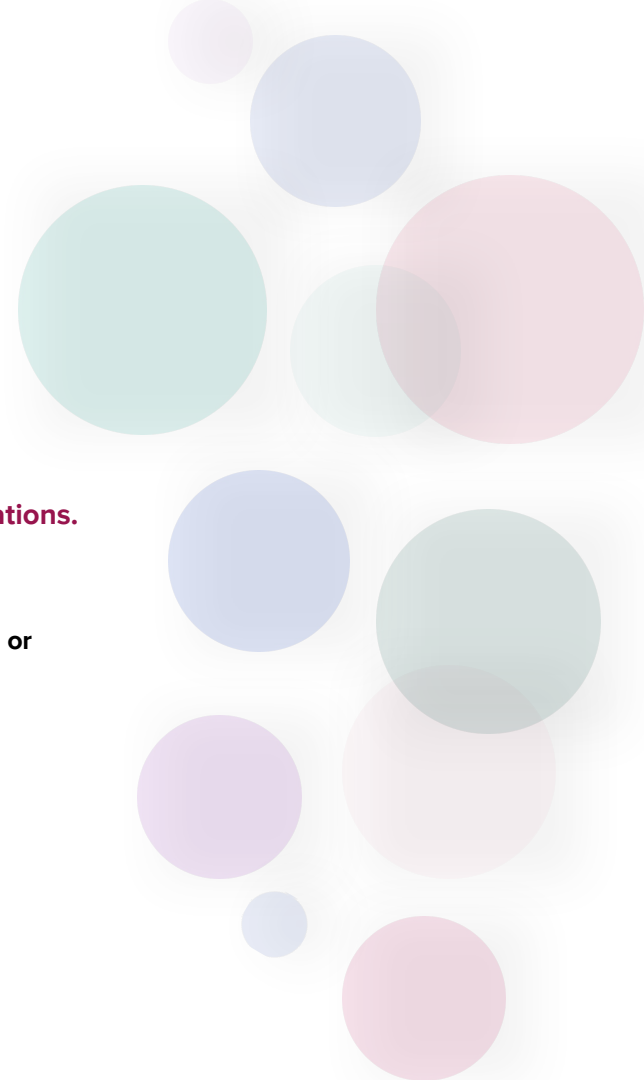
Major Insights and Findings

Note: Legends to be noted in the following pages	
All	9
Most	7-8
Some	4-6
Few	1-3

DISTINCTION OF COLORS



- Most users were able to identify the presence of seven color blocks in the data visualizations. However, they encountered difficulty in distinguishing all seven colors distinctively
- Possible reasons for this issue could be the **each block/slice size in the chart being too small or the colors used having low contrast ratios.**



COMPREHENSION OF COLORS



No	Colors	User 1	User 2	User 3	User 4	User 5	User 6	User 7	User 8	User 9
		Medium Intensity	Low Intensity	Low Intensity	Medium Intensity	Medium Intensity	Medium Intensity	Medium Intensity	High Intensity	High Intensity
1			Looks similar						Green	Green
2					Blue					
3										
4		RED		Pink	Pink		Grey			Grey
5		GREY, DARKER RED			Grey	Maroon	Grey		Dark brown	Dark red/ Dark Pink / Grey
6						Green			Green	Green
7				Pink & Light blue merging when further away	Green					

COMPREHENSION OF COLORS - SIMILAR COLORS



- User with medium intensity color blindness perceived Dark green as either Grey or Maroon
- Users with relatively low intensity did not face any challenge in perceiving Dark green color

→ Most users, regardless of their degree of color blindness, found it difficult to tell the distinguish between pink and light green. Particularly, those with deuteranopia tended to perceive light green as pink.



- Some Users with Protanomaly color blindness discerned Pink as Green



- Some users with medium to high intensity couldn't tell the difference between the hues of blue, red, and green as they thought the colors were relatively similar.
- Few users felt Dark green and light green looks similar

66 I can only see 4 colors here Dark pink and Dark green is similar Pink, Light green and dark green is similar- U5,
Deuteranopia, Medium intensity 99

COMPREHENSION OF COLORS - CONTRASTING COLORS



The majority of color-blind users, regardless of the level of color blindness, perceived Blue as the most prominent contrasting color, followed by Few perceived Purple, in both the Light theme and Dark theme.

- Some of the reason for the blue to prominent could be
 - Its high contrasting ratio compared to other 6 colors
 - High contrast ratio of blue with white and black background



Some users found the colors pink-magenta and blue-purple to exhibit a significant contrast in the **dark theme** compared to the light theme.

☞ Blue color is something i can easily distinguish from the entire colors blocks, then Purple also is pop out here - U1, Deuteranopia, High intensity ☞☞

☞ Strangely The colors like Magenda and Pink, Blue purple looks contrasting the dark theme as well as Green looks more Greener in Dark theme where in Light theme green looked more Grey- U5, Deuteranopia, Medium intensity ☞☞

Medium Intensity	Low Intensity	Low Intensity	Medium Intensity	Medium Intensity	Medium Intensity	Medium Intensity	High Intensity	High Intensity
User 1	User 2	User 3	User 4	User 5	User 6	User 7	User 8	User 9
Light theme			Light theme	Light theme	Light theme	Light theme	Light theme	
Very similar		Very similar		Very similar	Very similar			Very similar
			Very similar					
						Merging		
							Very similar	
Very similar	Very similar	Very similar	Very similar	Very similar				
				Very similar	Very similar	Very similar	Very similar	
								Contrasting
Very similar								
contrasting								
					Dark theme	Dark theme	Dark theme	Dark theme
	Colors look different in Dark theme				contrasting	contrasting	Contrasting	Very similar

Though users were able to identify most colors, a fair number of them expressed dissatisfaction with the color palette, describing it as dull and noting that certain colors were too similar, making them difficult to differentiate.

- This could be due to the factors like
 - **Size of the block/slice the color is occupied**
 - **Low Contrast ratios between colors from the perspective of color blind users**

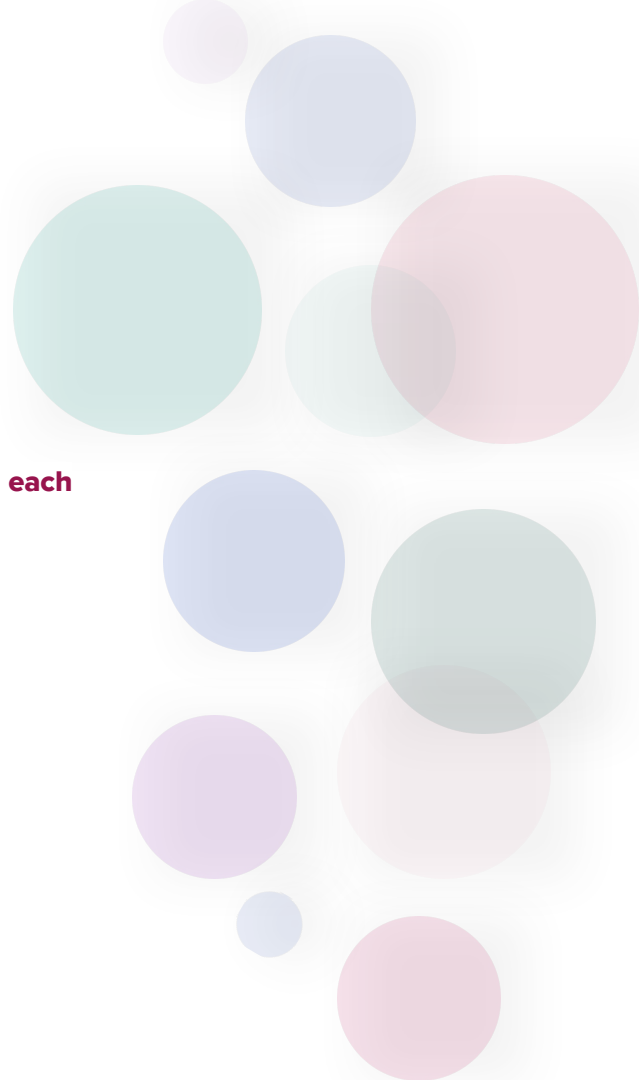
BB Colors are good enough but personally i like to see vibrant colors to make my life easier - U3, Deuteranopia, low intensity??

BB Colors are looking dull. It should be more contrasting you could use colors like blue, Yellow, Orang, brown more like desaturated red etc will help in distinguishing colors- U4, Protanopia, Medium intensity ??

RECOMMENDATIONS



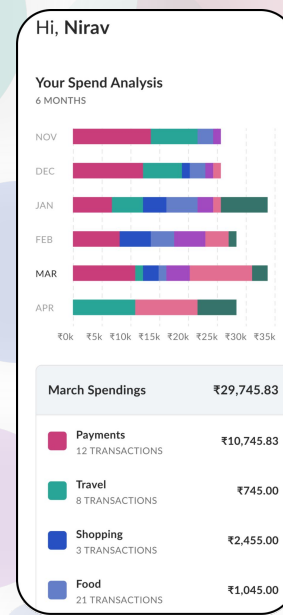
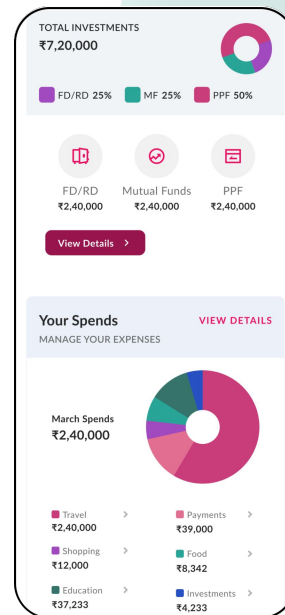
Almost all Users highly recommended to use colors that are vibrant and contrasting to each other such as **Yellow, **Orange**, **Bright Red** and **Blue**, etc.**



COMPREHENSION OF CHARTS

- It was observed that users' preferences for charts varied depending on how they interpreted data visualizations.
- Most users said **Pie chart/ donut chart** was most **preferred way of view data visualisation** as they are **accustomed to viewing the similar representation**
 - Most users said **vertical chart** was **easy to figure the 7 blocks** in the bar as it **was separated by white line**. However **vertical charts** was the **least preferred** one among all the charts as user felt its **complicate to distinguish the colors that are in narrow blocks**
 - Some users perceived **horizontal bar charts** as providing **better color visibility** since the **width of each color block** was **sufficient for clear differentiation**.

Since the colors are separated by white line it's clearly understanding that there are 7 set of colors- U2, Protanopia, low intensity



BACKGROUND THEME

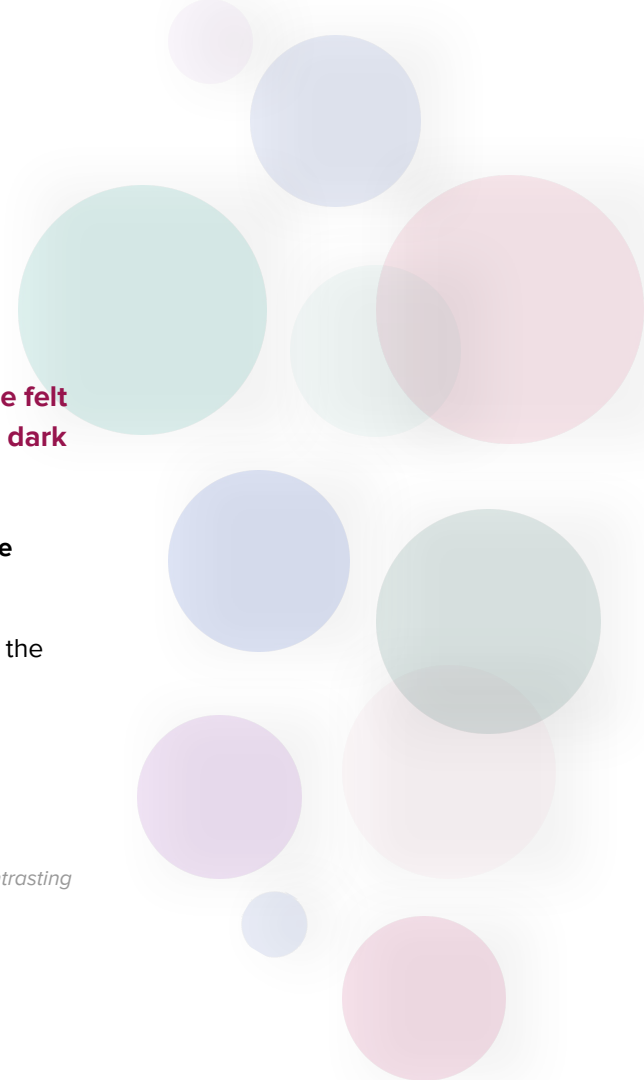


→ **Preference of light theme/ Dark theme found to be very subjective among users as some felt Dark theme is popping out all the colors while others felts Colors are submerging in the dark theme compared to white theme**

- 3/9 Users with **medium to high intensity** majorly found the **certain colors to be merging in the light background**
- 3/9 users with **high intensity** felt the **white theme is more comfortable for the eyes** to look at the colors

☺☺ *White theme is more comforting to the eyes to look at the colors, In dark theme colors are popping out causing discomfort to the eyes- U9, Deuteranopia, High intensity* ☹☹

☺☺ *In Dark theme Color standing out light blue and purple is similar Dark pink and light pink is contrasting*
- U6, Deuteranopia, Medium intensity ☹☹



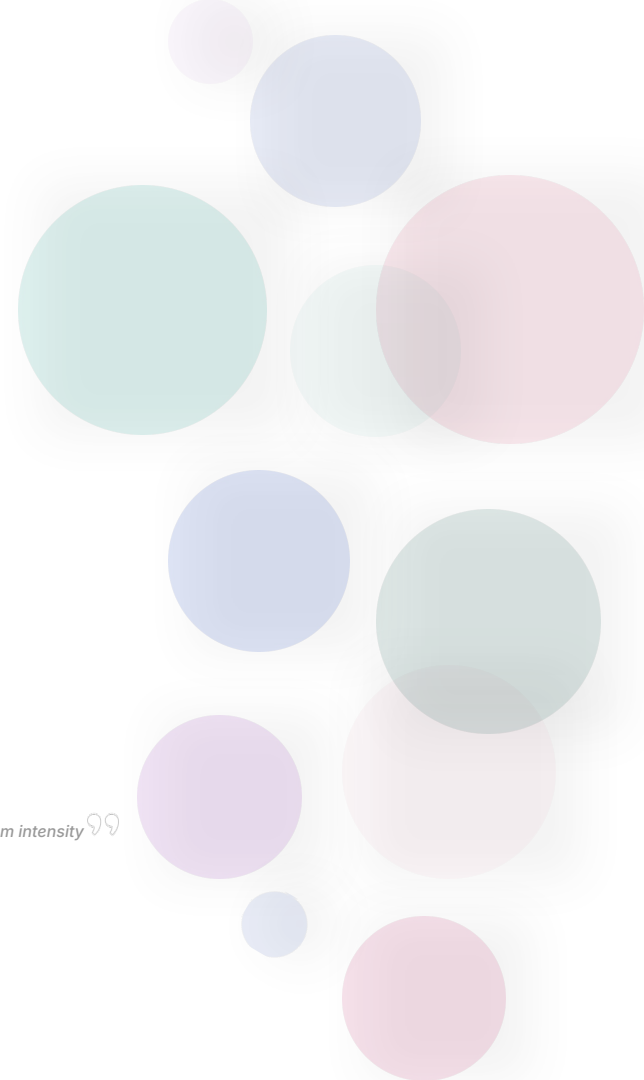
ASPECTS THAT CONTRIBUTE TO THE DISTINCTION OF COLORS



- **Users often rely on additional elements/factors such as the respective legends' values and iconography to differentiate colors in data visualizations.**
 - **Elements/Aspects such as**
 - **size of the graph, Legend**
 - **Distinction line between colors**
 - **Icons**
- **Some users encountered difficulty in identifying the colors of legends due to the small size of the legend itself.**

👁👁 Icons don't add much value but Size of the element matters when it comes to identifying the colors
- U2, Deuteranopia, Low intensity 👁👁

👁👁 Use logic, looks at the values and icons, its shape and form to distinguish- U1, Deuteranopia, Medium intensity 👁👁

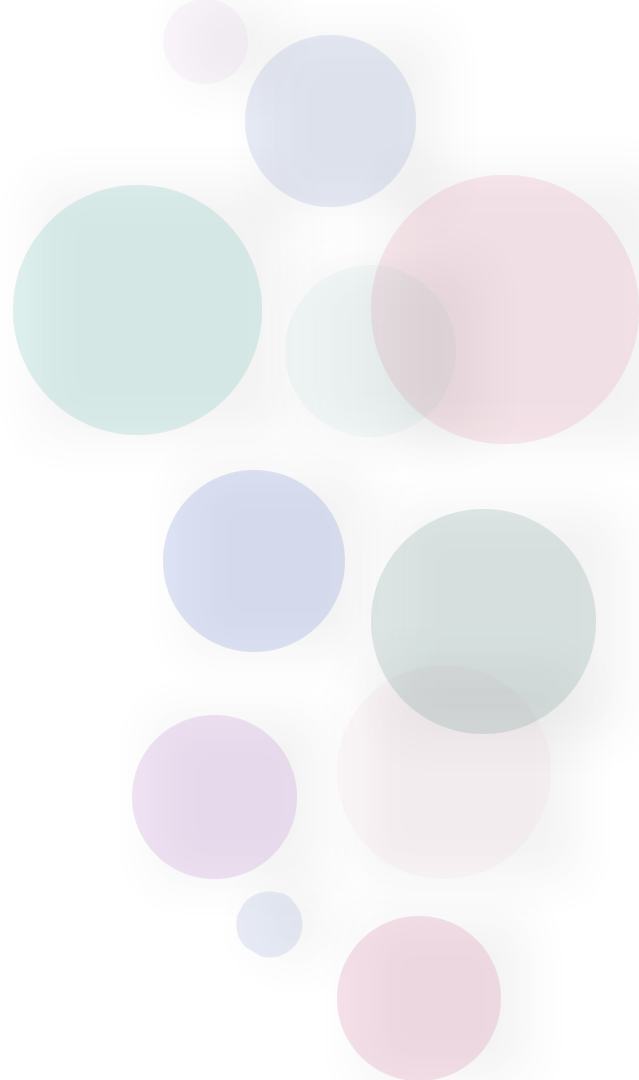


RECOMMENDATIONS



Few users suggested the **inclusion of interactive features in data visualizations to enhance interpretation and improve the overall user experience.**

- **Introducing interactive elements or touch points** within the bar chart would allow users to **directly interact with specific slices or blocks**, enabling them to **obtain relevant information or values for a particular segment.**



SITUATION BEHAVIOR



- As the Gaming platform colorblindness mode, Users often change the colors like red and green to vibrant colors like yellow or blue
- In trading applications, some users opt to customize the color of candlestick bars according to their preferences in order to enhance visibility and distinguishability.

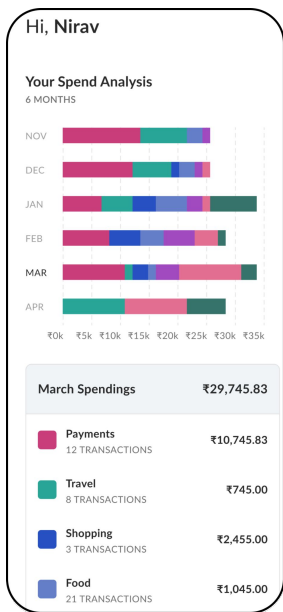
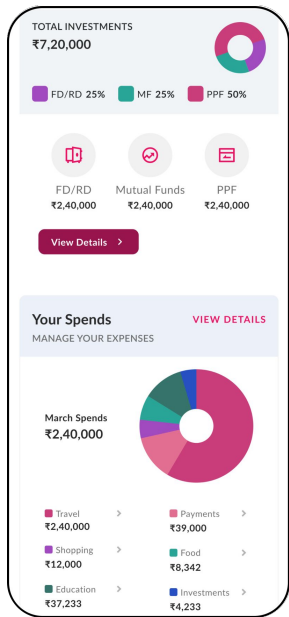


RECOMMENDATIONS



Enabling users to **customize the color palette based on their preferences** has the potential to **enhance the interpretation of data visualizations for all users**. This customization option would **promote inclusivity and cater to the unique vision abilities of each individual**.





Q & A

THANK YOU

